Covariant superstrings and compactification

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Introduction
- The role of string theory
- How to use it?
- The need for covariant superstrings

Superstrings

Covariant superstrings

Hybrid strings
- Flux deformed Calabi-Yau compactification

Pure spinor strings

Conclusion

Tools *versus* Physics

The role of string theory is as a calculational tool for quantum field theories coupled to gravity.

TOOL	$\longleftrightarrow$	PHYSICS
Calculus		classical mechanics
Functional Analysis		quantum mechanics
QFT		Fundamental fields with spin ≤ 1
Strings		Fundamental fields with spin ≤ 2
$\vdots$		$\vdots$

Note increasing symmetry and decreasing generality.

Compactification and Holography

- Critical superstrings live in 10 flat dimensions
- We live in 4 (almost flat) dimensions
- To use our tool, we need to reduce $10 \to 4$
- Two complementary possibilities:
- Compactification
- Holography

Fluxes and branes

Both approaches to dimensional reduction require analogue of electrons and/or background electromagnetic flux.

Compactification

- Compactifications on good string backgrounds without flux have moduli.
- Example: Heterotic string on Calabi-Yau 3-fold has $N = 1$ supersymmetry $\Rightarrow$ flat directions.

Holography

- Holographic construction: Branes $\rightarrow$ Flux.

Fluxes and branes

Both approaches to dimensional reduction require analogue of electrons and/or background electromagnetic flux.

Compactification

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- Example: Heterotic string on Calabi-Yau 3-fold has $N = 1$ supersymmetry $\Rightarrow$ flat directions.

Holography

- Holographic construction: Branes $\rightarrow$ Flux.
- Example: To get $N = 4$ super-Yang-Mills we start with a stack of many D3-branes at weak string coupling. Increasing the coupling gives $AdS_5 \times S^5$ with 5-form flux on the S^5.

Superstrings

Superstring = Bosonic string + SUSY
Bosonic string $x : \Sigma \to M$

$$S \sim \int (\partial x)^2$$

Supersymmetry in field theory

$$\left.\begin{aligned} \delta\phi &= \epsilon\psi \\ \delta\psi &= \bar{\epsilon}\not{\partial}\phi \end{aligned}\right\} [\delta, \delta] \sim \bar{\epsilon}\gamma^m\epsilon\partial_m$$

Superstrings

Supersymmetry in string theory

world-sheet	space-time
$\delta x^m = \epsilon\psi^m$	$\delta x^m = i\epsilon\gamma^m\theta$
$\delta\psi^m = \bar{\epsilon}\partial x^m$	$\delta\theta = \epsilon$

Open superstrings

RNS$\pm$ spin structure:	GS$\pm$ spin structure:
(anti-)periodic B.C.	space-time (anti-)Weyl
$S \sim \int (\partial x)^2 + \psi\partial\psi$	$S \sim \int \Pi^2 + S_{\mathrm{WZ}}$
	$\Pi^m = \mathrm{d}x^m + i\theta\gamma^m\mathrm{d}\theta$

Superstrings

Closed superstrings

	left-moving	right-moving	
SUSY	$1\pm$	0	Heterotic O/E
	$1\pm$	$1\pm$	Type II A/B

Superstrings

Hidden symmetries

RNS	GS
Spacetime SUSY	Worldsheet Siegel symmetry
$\{\psi_0^m, \psi_0^n\} = 2g^{mn}$	$\delta\Pi^m = \mathrm{d}\bar{\theta}\gamma^m\delta\theta$ $\delta\theta = \not{\Pi}\kappa$
vacuum: spacetime spinor	generated by $\not{\Pi}d$

Superstrings

Hidden symmetries

RNS	GS
Bosonization $\bar{\psi}\psi \sim i\partial H$	Light-cone gauge
$(\psi_0^m \pm J^m{}_n \psi_0^n) = \mathrm{e}^{i(\pm H_0 \pm \cdots \pm H_4)}$	$\gamma^+\theta = 0$
$\downarrow$ GSO-projection	$\Downarrow$
$\Theta_{\mathbf{s}} = \mathrm{e}^{i\sum_a s_a H_a}$	x^i , S^a (triality)

Superstrings

Benefits/drawbacks

RNS	GS
Super-Virasoro	Unitary gauge
$\psi\Theta \sim \frac{1}{\sqrt{z}} \dots$	$p_\theta = \frac{\delta S}{\delta \dot{\theta}} \sim i\bar{\theta}\gamma^m \partial x_m + \dots$

Superstrings

Quantization

RNS	GS
Super-Virasoro constraints 1st class	Mixed 1st and 2nd class
Covariant quantization	Non-covariant quantization
Trivial Ramond backgrounds	Trivial NS backgrounds

Neither possible for mixed backgrounds.

Covariant superstrings

Origin of problem in covariant quantization of GS string:
2nd class constraints from composite θ-momentum
Siegel "solution": Include fundamental θ-momentum

$$\int (\partial x)^2 \quad \rightarrow \quad \int (\partial x)^2 + p_\theta \partial\theta$$

Manifestly spacetime supersymmetric form

$$S_{\text{Siegel}} = S_{\text{GS}} + \int d\partial\theta$$

Siegel constraints $\qquad d = p_\theta + i\bar{\theta}\gamma^m \partial x_m + \theta\theta\partial\theta\text{-term}$

Covariant superstrings

Operator products $\quad d_\alpha(z) d_\beta(0) \sim \frac{1}{z}(\gamma^m)_{\alpha\beta}\Pi_m(0)$

Still mixed 1st and 2nd class since $T = \Pi^m \Pi_m + \ldots$ is 1st class.

Two solutions to the problem of separating constraints due to Berkovits:

1. Hybrid strings
2. Pure spinor strings

Hybrid strings

Consider RNS in $M = \mathbb{R}^4 \times \mathbb{C}^3$ background

	"spacetime"	"internal"
matter	x^m, ψ^m	$y^i, \bar{y}^{\bar{\imath}}, \psi^i, \bar{\psi}^{\bar{\imath}}$
ghost	b, c, β, γ	

Berkovits	$\downarrow$	$\downarrow$
variables	$x^m, p_\alpha, \theta^\alpha, \bar{p}_{\dot{\alpha}}, \bar{\theta}^{\dot{\alpha}}, \rho$	$y^i, \bar{y}^{\bar{\imath}}, \psi^i, \bar{\psi}^{\bar{\imath}}$

Examples

$$
\begin{aligned}
(\theta^\alpha) &\sim \exp \frac{i}{2}\left(-i\phi \pm (H_0 + H_1) - H\right) \\
(\bar{\theta}^{\dot{\alpha}}) &\sim c\xi \exp \frac{i}{2}\left(3i\phi \pm (H_0 - H_1) + H\right)
\end{aligned}
$$

Hybrid strings

Canonical operator products

$$y^i(z,\bar{z})\bar{y}^{\bar{\jmath}}(0) \sim \delta^{i\bar{\jmath}} \log|z|^2 \ , \ \psi^i(z)\bar{\psi}^{\bar{\jmath}}(0) \sim \frac{\delta^{i\bar{\jmath}}}{z} \ , \ \rho(z)\rho(0) \sim -\log z$$

$$x^m(z,\bar{z})x^n(0) \sim \log|z|^2 \ , \ p_\alpha(z)\theta^\beta(0) \sim \frac{\delta_\alpha^\beta}{z} \ , \ \bar{p}_{\dot{\alpha}}(z)\bar{\theta}^{\dot{\beta}}(0) \sim \frac{\delta_{\dot{\alpha}}^{\dot{\beta}}}{z}$$

Super-Virasoro constraints

$$\begin{aligned} T &= \Pi^m\Pi_m + d\partial\theta + \bar{d}\partial\bar{\theta} + \delta_{i\bar{\jmath}}\partial y^i\partial\bar{y}^{\bar{\jmath}} + \delta_{i\bar{\jmath}}\psi^i \overleftrightarrow{\partial} \bar{\psi}^{\bar{\jmath}} \\ G^+ &= \mathrm{e}^\rho d^2 + i\delta_{i\bar{\jmath}}\psi^i\partial\bar{y}^{\bar{\jmath}} \\ G^- &= \mathrm{e}^{-\rho}\bar{d}^2 + i\delta_{i\bar{\jmath}}\bar{\psi}^{\bar{\jmath}}\partial y^i \\ J &= -\partial\rho - \delta_{i\bar{\jmath}}\bar{\psi}^{\bar{\jmath}}\psi^i \end{aligned}$$

Hybrid strings

All constraints first class with commuting factors

$$\mathrm{SCA}_{10} = \mathrm{SCA}_4 \oplus \mathrm{SCA}_6 \quad \Rightarrow \quad \mathrm{Rep}_{10} = \mathrm{Rep}_4 \otimes \mathrm{Rep}_6$$

$$\mathrm{Rep}_4 \quad \Leftrightarrow \quad \Phi(x, \theta, \bar{\theta}, p, \bar{p}, \rho)$$

$$\mathrm{Rep}_6 \quad \Leftrightarrow \quad \sum \Omega_{i_1 \ldots i_p \bar{\jmath}_1 \ldots \bar{\jmath}_q}(y, \bar{y}) \psi^{i_1} \ldots \psi^{i_p} \bar{\psi}^{\bar{\jmath}_1} \ldots \bar{\psi}^{\bar{\jmath}_q}$$

Physical state conditions

Physical state conditions for unintegrated vertex operators are the usual RNS conditions. No double poles with Virasoro implies conformal weight 0. First possibility: Ω has weight $0 \Rightarrow \Omega = 1 \Rightarrow \Phi$ function of 0-modes of $x, \theta, \bar{\theta}$ only $\Rightarrow \Phi$ is a *superfield* and

$$d(z)\Phi \sim \frac{1}{z} D\Phi \quad , \quad \bar{d}(z)\Phi \sim \frac{1}{z} \bar{D}\Phi$$

$$D = \frac{\partial}{\partial\theta} + i(\bar{\theta}\sigma^m)\frac{\partial}{\partial x^m} \quad , \quad \bar{D} = -\frac{\partial}{\partial\theta} - i(\sigma^m\theta)\frac{\partial}{\partial x^m}$$

Superspace covariant derivatives with algebra

$$\{D, D\} = 0 \ , \ \ \{D, \bar{D}\} = -2i\sigma^m\partial_m \ , \ \ \{\bar{D}, \bar{D}\} = 0$$

(Anti-)chiral fields

(Anti-)chiral fields:

No single pole with (G^+) G^-:

$$0 = \oint G^- U = \left(\oint \mathrm{e}^{-\rho} \bar{d}^2(z)\Phi(0)\right)\Omega(0) + \Phi(0)\left(\oint \bar{\psi}\partial y(z)\Omega(0)\right)$$
$$= \left(\mathrm{e}^{-\rho}\bar{d}\bar{D}\Phi\right)\Omega + \Phi\left(\bar{\psi}\frac{\partial\Omega}{\partial\bar{y}} + \partial y\frac{\delta\Omega}{\delta\psi}\right)$$

$\Rightarrow$ Φ is a chiral superfield and Ω is independent of ψ.
$\Rightarrow$ $\Omega = \omega(y,\bar{y})_{\bar{\imath}_1\ldots\bar{\imath}_q}\bar{\psi}^{\bar{\imath}_1}\ldots\bar{\psi}^{\bar{\imath}_q}$ and $\bar{\partial}\Omega = 0$
$\Rightarrow$ Ω holomorphic $(0,q)$-form!

"Chiral ring" of "internal" space $\Leftrightarrow$ cohomology ring.

(Anti-)chiral fields

Spin-$\frac{3}{2}$ generators

Correlated	$G^+ = G_4^+ + G_6^+,$	$G^- = G_4^- + G_6^-$	and $J = J_4 + J_6$
or			
Anti-corr.	$G^+ = G_4^+ + G_6^-,$	$G^- = G_4^- + G_6^+$	and $J = J_4 - J_6$

Spacetime chiral superfields correlated with internal chiral ring
or
spacetime anti-chiral superfield correlated with chiral ring and
vice versa.

Calabi-Yau compactification

Type IIB string on Calabi-Yau X
Left- and right-moving SCAs (Type II)
Both correlated (Type B)

	(chiral,chiral)	(chiral,anti-chiral)
spacetime	$N = 2$ chiral	$N = 2$ twisted-chiral
multiplet	vector	hyper
cohomology	$H^{2,1}_{\bar\partial}(X)$	$H^{1,1}_{\bar\partial}(X)$
deformation	$\mathbb{C}$-structure	Kähler-structure

Calabi-Yau compactification

Vector-multiplets (chiral,chiral) $M = \sum_{p=1}^{h^{2,1}} M^p \Omega^p_{(-1,-1)}$

$$M^p = \phi^p + \theta_L^2 D^p_{++} + \theta_R^2 D^p_{--} + \theta_L \theta_R D^p_{+-} + \theta_L \sigma^{mn} \theta_R F^p_{mn} + \dots$$

Integrated vertex operator

$$I^p = \int G_L^+(G_R^+(M^p)) + \text{h.c.} = \int \mathrm{e}^{\rho_L + \rho_R}(d_L d_R) D^p_{+-} + \text{h.c.} + \dots$$

Flux deformation

D3-brane background preserves $\theta^+ = \theta_L + i\theta_R$ spacetime supersymmetry

$$M^p = \phi^p + (\theta^+)^2 X^p_{++} + (\theta^-)^2 X^p_{--} + (\theta^+\theta^-) X^p_{+-} + \dots$$

Flux deformed background preserving D3 symmetry (Klebanov-Strassler)

$$X^p_{--} = 0 = X^p_{+-} \qquad X^p_{++} \sim g_s \bar{G}^p \quad \text{(ISD flux)}$$

Flux deformation

Vertex operator for flux deformation

$$I_{\mathrm{D3}} = \sum_{p=1}^{h^{2,1}} I^p \Omega^p = \int G_L^+(G_R^+(M^p)) + \mathrm{h.c.}$$
$$\sim g_s \int \mathrm{e}^{\rho_L + \rho_R}(d_L d_R)\bar{G}^p \Omega^p + \mathrm{h.c.}$$

Conformal symmetry breaking divergence

$$I_{\mathrm{D3}} I_{\mathrm{D3}} \sim g_s^2 \log \Lambda \int \Pi_L \Pi_R |G|^2$$

Cancel with *spacetime* gravitational vertex operator.

Warping

In flat space

$$U_1 \sim (\theta_L \theta_R)(\bar{\theta}_L \bar{\theta}_R) A + \ldots \quad \Rightarrow \quad I_1 = \int \Pi_L \Pi_R A + \ldots$$

Background field expansion

$$I_1 \to \int \Pi_L \Pi_R y^i \bar{y}^{\bar{\jmath}} \nabla_i \bar{\nabla}_{\bar{\jmath}} A \sim \log \Lambda \int \Pi_L \Pi_R \Delta A$$

At 1-loop therefore $\boxed{\Delta A \sim g_s^2 |G|^2}$

Warp factor of the spacetime metric ([5] with Brenno Carlini Vallilo).

Significance

The results of the work with Brenno Carlini Vallilo have many implications:

- This approach allows a worldsheet (*i.e.* string-theoretic) derivation of many spacetime effects of background fluxes
- Can in principle be used to find all effects such as torsion, non-commutativity, *et cetera*
- Suggests the validity of the flux background as a string background due to existence of worldsheet SCA (no breaking of worldsheet supersymmetry)
- Allows computation of scattering amplitudes in non-trivial backgrounds
- Existence of new type of mirror symmetry for heterotic strings?

Unfortunately, "picture-changing problem" persists...

Pure spinor strings

In ten dimensions we can introduce spinor harmonics to separate constraints

$$\{d_\alpha, d_\beta\} = -2i(\gamma^m)_{\alpha\beta}\Pi_m \quad \rightarrow \quad \left\{\lambda^\alpha d_\alpha, \lambda^\beta d_\beta\right\} = -2i\,(\lambda\gamma^m\lambda)\,\Pi_m$$

Can be chosen to vanish $\quad \Leftrightarrow \quad \lambda\gamma^m\lambda = 0$
Fierz identity

$$\lambda^\alpha\lambda^\beta = \sum_{p \text{ odd}} (\gamma_{[p]})^{\alpha\beta}(\lambda\gamma^{[p]}\lambda) = (\gamma_{m_1\ldots m_5})^{\alpha\beta}(\lambda\gamma^{m_1\ldots m_5}\lambda)$$

defines self-dual 5-plane
$\Leftrightarrow$ Lagrangian sub-bundle of $T^{\mathbb{C}}M$
$\Leftrightarrow$ "pure spinor"

Pure spinor strings

Berkovits operator

$$Q = \oint \lambda^\alpha d_\alpha \quad \Rightarrow \quad Q^2 = 0$$

Cohomology at ghost number 1

$$U = \lambda^\alpha A_\alpha(x,\theta) \in H^1_Q \quad \Rightarrow \quad U = \Gamma^m A_m + \Gamma_\alpha \chi^\alpha - Q(G)$$

where $\Gamma^m = \lambda\gamma^m\theta$ and $\Gamma_\alpha = -\frac{1}{2}(\theta\gamma^{mpq}\theta)(\lambda\gamma_{mnq})_\alpha$ span the 0-momentum cohomology at ghost number 1.
Cohomology corresponds to a photon A_m and photino χ^α.

Pure spinor strings

Applications

- Compactification and relation to hybrid
- Holographic backgrounds
- ...

Open problems

- Looks like gauge fixing of a superstring with some fermionic symmetry. *Which string ??*
- Spectrum
- ...

In other news...

Hybrid

- Compactification on $K3$, $K3 \times T^2$ [2], CY_3, CY_4, ...
- Flux compactification on CY [1, 3], superpotentials, non-commutativity, ...
- Superstring field theory, black holes, amplitudes, ...

Pure spinor

- Compactification on CY [4]
- $AdS_5 \times S^5$ background and possible proof of Maldacena duality
- Integrability in the $AdS_5 \times S^5$ sigma model [5]

Fin

Thank you!

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